



in Thailand

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Demand for Collaborative Computing in Thailand



- Collaborative computing is increasingly important for many reasons
 - Allow the access to human expertise that distributed across geographically distance
 - Allow large work group to solve a complex problem together
 - Reduce the time and expense used to travel



Potential Applications



- Tele education, E-learning
- Tele medicine
- Technical discussion
- Crisis management
- Remote conference , job interview
- Trade negotiation



The interest in Access Grid System



- Start at Kasetsart University
 - 2001 Setting up an experimental system and successfully linked with
 - MCS, Argonne National Laboratory
 - University of Sydney
 - KMIT North Bangkok
 - 2002/2003 Building a full access Grid node
 - Successful test with NUS, U. of Sydney, AIST, KU, KMITNB, QUT
- Other sites
 - Experimental system exists at KMITNB, SUT
 - Our Mobile Unit (Single node AG)
 - Potential new AG node : AIT, NECTEC





KU Access Grid



- Hardware
 - Audio Digitizer System
 - Video Capture System
 - Multiple Video Capture Board (3 board)
 - Display System
 - Dual Head VGA Display Card (Matrox Graphics Card)
- Audio Equipment
 - Microphone, Mixer, Amplifier
 - Echo canceller (Gentner Echo cancellor Microphone)
- Access grid Software
- Access Grid Toolkit 2.1







Problem



- Social and Economics Issues
 - The awareness of the technology is still very low.
 - The initial implementation cost is relatively high compared to Thai economy
- Technical issues
 - IP multicast in Thailand is not available everywhere. Bridge are used for the testing (under the work of solving it between KU, Uninet). Recently, multicast started to work for some site such as NECTEC, AIT which has no AGnode
 - Bandwidth to Thailand is inadequate and shared, no dedicated research network. Hard to test any advanced technology.
 - Equipment is difficult to purchase since vendor like to sell extremely expensive equipment or solution to Thai people.
 - AG software is relatively hard to setup and operate



Opportunities



- Access grid can be used in
 - Campus or corporate environment since bandwidth and multicast is not a problem
- Ideas
 - Remote training between branches
 - Remote meeting, live
 - Remote interview
 - Education, link multiple room together
- Where is the killer app?



Planned Activities



- Virtual Research Seminar (VRS)
 - Join seminar among a multiple research group
 - Testing with Chulalongkorn university
- Setup Campus AG infrastructure
 - E-Learning
 - Public Virtual meeting Space
- SCGlobal 2003 participation
- Workshop, training



Summary



- Access Grid is an important technology
 - Cost effective
 - Multipoint
 - Realtime
 - Bandwidth Efficient (IP multicast based)
- Access grid can used in Thailand for many applications.
- Many problems exist that need to be solve to stimulate the adoption of this technology in Thailand



The end